Selection of Monochromator Crystals for Pb L-, Br K-, and Ag K-edge XANES Using a Laboratory-Based Spectrometer

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Abstract

We recorded Pb L3-, Br and Ag K-edge XANES spectra using conventional laboratory-based spectrometer using various monochromator crystals: Si(400), Si(620), Si(800), Si(12 4 0), Ge(840) and Ge(880). Undesirable low-index reflections from the single crystal monochromator and the characteristic X-ray lines were suppressed using appropriate thick aluminum filter, and the corresponding optical setting were discussed. The absorption edge of various Pb compounds at Pb L3-edge was evaluated using two methods: E0-half, defined as the position at the half-height of normalized XANES, and E0-diff, defined as the peak position of the first derivative of XANES. The influence of energy resolution on the apparent absorption edge and the resulting chemical shift were discussed. The use of a monochromator crystal with small d-spacing resulted in XANES spectra of higher resolution, comparable to those recorded at synchrotron radiation facilities. The energy resolution did not affect the apparent chemical shift derived from E0-half, but significantly affect the shift evaluated from E0-diff. The first derivative of Pb L3-edge XANES spectra with lower energy resolution is suggested to be preferable for oxidation state analysis of unknown samples. The application of high-order reflections Ge(880) at Ag K-edge lab-XAFS provided XANES spectra, which are applicable for chemical state analysis.

1. Introduction

X-ray absorption experiments are typically conducted using high-intensity X-rays from synchrotron radiation (SR) sources. X-ray absorption fine structure (XAFS) experiments had been developed using laboratory-based spectrometer based on X-ray tubes, and until the early 1990s, many studies focused on the development and the utilization [1-5]. Laboratory XAFS (lab-XAFS) offers advantages such as reduced constraints on travel to SR facilities, and the shorter waiting times for allocated beamtime, enabling more efficient execution of sample preparation, evaluation, and improvement in research and development work. Although lab-XAFS has limitations in terms of applicable samples and experimental details due to its lower incident beam intensity compared to SR sources, it is possible to obtain results comparable to those from synchrotron experiments by optimizing energy resolution, photon flux, and measurement time according to the instrument characteristics. Notably, the extremely low risk of radiation damage is a significant advantage of lab-XAFS experiment. Recent development in novel optical systems, X-ray detectors, and compact X-ray sources have led to a renewed increase in research on laboratory-based XAFS, establishing it as a practical and complementary approach to synchrotron-based measurements [6-14].

Other problems needed to overcome for typical lab-XAFS experiment are suppression of undesired reflection from low index planes of a monochromator crystal, as well as the contamination from characteristic lines of the anode. To address this issue, Udagawa et al. had developed a double-crystal spectrophotometer that combines two crystals differing in either materials or index plane [4, 15]. In cases where X-ray detectors with limited energy resolution are used in a typical single-crystal spectrometer, appropriate filters must be inserted and the detector settings carefully adjusted. In the present study, we conducted XAFS experiments in the 13–25 keV energy range using a conventional laboratory-based spectrometer. Various configurations of monochromator crystals, optical systems, and detector settings were tested to identify suitable measurement conditions. We also compared XANES and EXAFS spectra obtained under these different setups.

2. Experimental

X-ray absorption experiments were conducted using a laboratory-type spectrometer R-XAS Looper (Rigaku) [16] equipped with an open-type demountable X-ray tube generator (XG) under air. The W filament and W anode, a Xe-gas sealed proportional counter (SPC), and a scintillation counter (SC) were used as the X-ray source and detectors for incident and transmitted X-rays, respectively. The accelerating voltage of XG and the tube current for XAFS experiments at the Pb L_3-, Br, and Ag K-edges were typically 27 kV–50 mA, 30 kV–50 mA, and 40 kV–50 mA, respectively. A Johann- or Johannson-type monochromator crystal with a Rowland circle radius of 320 mm is mounted on the monochromator. Photos of the spectrometer and the schematic diagram are shown in Fig. 1. The utilized curved monochromator crystals were Si(400), Si(620), Si(800), Si(12 4 0), Ge(840) and Ge(880), with the corresponding the lattice spacing (*2d*) of 2.715, 1.718, 1.358, 0.857, 1.265 and 1.000 Å. The monochromator crystal Si(800) and Si(12 4 0) are the same device as the Si(400) and Si(620), respectively.

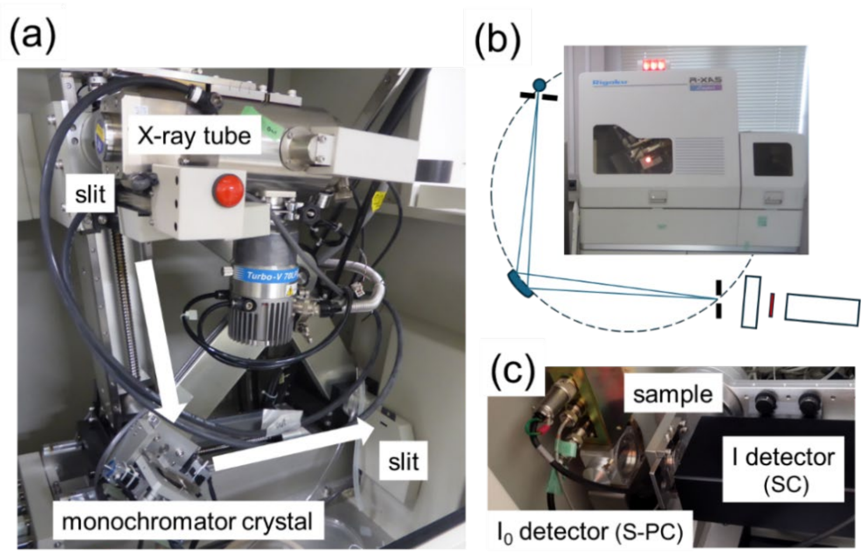

Fig. 1. The laboratory XAFS spectrometer and the schematic diagram.

The incident X-ray from the Si(800), Si(12 4 0) or Ge(880) monochromator contains undesired lower-order reflections, the intensities of which are much higher than those for the required beams. In cases Ge-based monochromator crystal was used and the X-ray tube voltage was 12 kV or higher, characteristic Ge K-lines (10-11 keV) are also contaminated in the incident X-ray. Then, appropriate thickness of aluminum filter made by folding of 15 µm foil was inserted between monochromator crystal and Xe-SPC (I0) detector to reduce undesirable X-rays with lower energy. Fig. 2 shows typical pulse height spectra of the Xe-SPC obtained using three types of monochromator crystals, with and without an Al filter. The pulse-height window of the SPC was set to 1.0 – 1.8 V. The thickness of the filter and the windows of pulse-height were determined by checking the profiles.

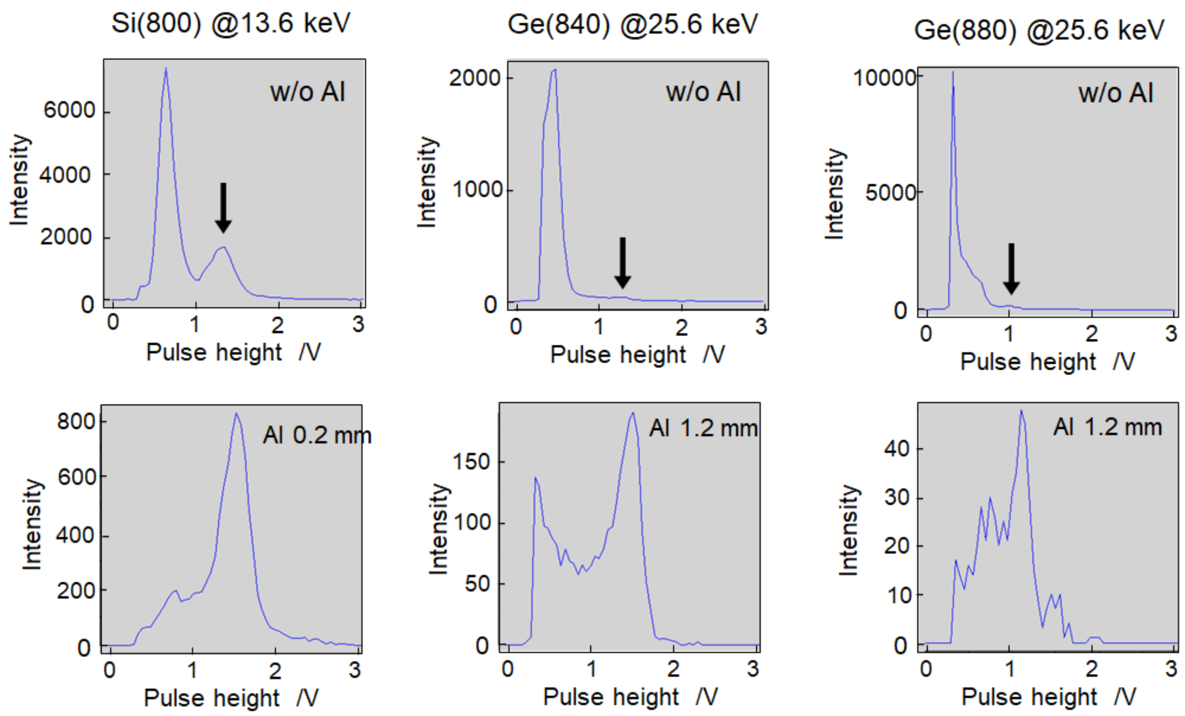

Fig. 2. Pulse height spectra of Xe-gas sealed proportional counter using Si(800), Ge(840) and Ge(880) monochromator crystal in the presence or absence of Al filter. Peaks indicated by arrows are due to X-rays of the target energy.

The X-ray absorption spectra were recorded in a transmission mode at room temperature. For the XANES, background removal and normalization of the spectra were performed using the Igor Pro 6.2 program [17] according to the procedure reported by Tanaka et al. [18]. Energy was not calibrated. Data reduction for the EXAFS analysis was performed with the REX2000 program [19].

3. Results and discussion

First, we present the effects of the filer thickness on laboratory X-ray absorption spectroscopy at Ag K-edge using a Ge(880) monochromator crystal. Fig. 3 shows Ag K-edge XANES spectra of Ag foil recorded with different thickness of an Al filter. Without Al-filter, distinct two negative peaks are present in the absorption spectrum at 25560 and 25610 eV. The peaks disappeared in cases the used filter was more than 0.2 mm. The Bragg angles of these peaks correspond to 6390 and 6400 eV for diffraction lines from Ge(220), and these energies coincide with the Fe $K\alpha_1$ and $K\alpha_2$ lines. The edge-jump increased with the filter-thickness. The detected X-ray intensity for the scintillation counter (SC, I-detector) decreased monotonically with increasing filter thickness. In contrast, the intensity for the proportional counter (PC, I_0-detector) increased from no filter up to a thickness of 0.2 mm, and then decreased at larger thicknesses.

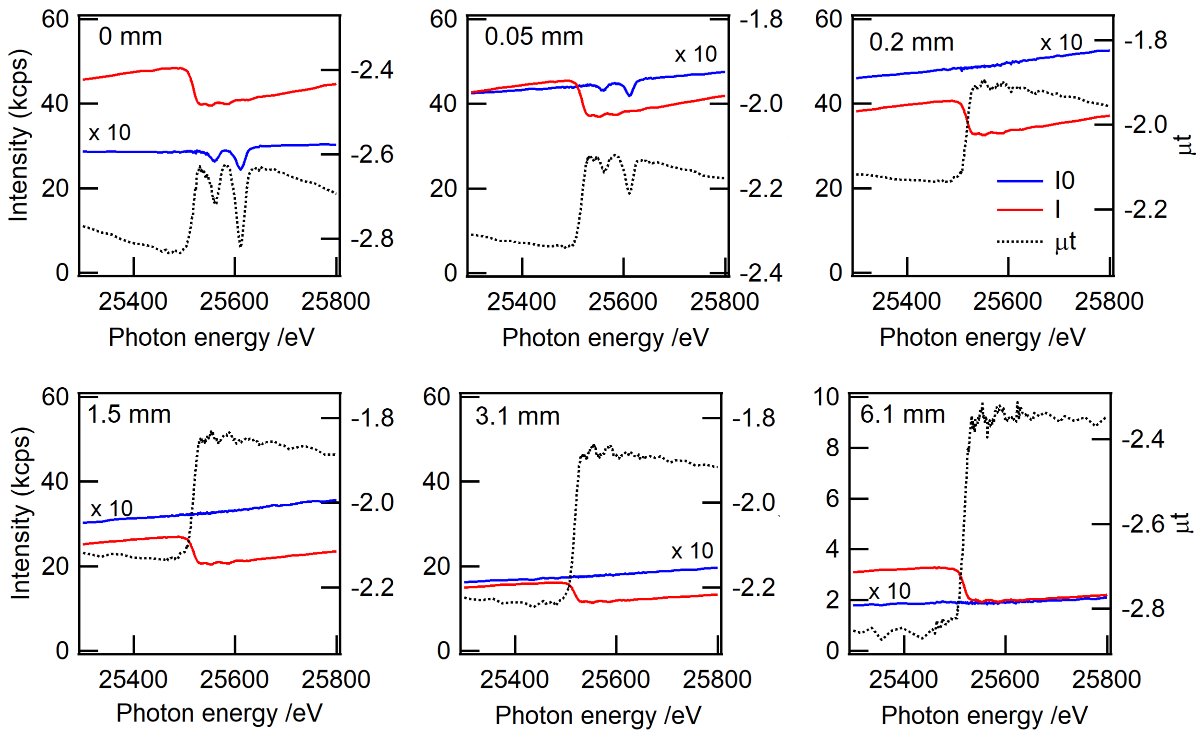

Fig. 3. Ag K-edge X-ray absorption spectra of Ag foil using various thickness of an Al filter. Monochromator crystal: Ge(880). Acquisition time: 50 sec/pt.

Similar studies were performed using Ge(840) and Si(12 4 0) monochromator crystals. The results are summarized in Fig. 4. The detected PC intensity decreased with increasing filter thickness for Ge(840) and Si(12 4 0), and the general trends trend differed from that for Ge(880). The undesirable incident X-rays from a Ge(840) or Si(12 4 0) monochromator crystal are Ge characteristic lines and low-order reflections from Si(620), respectively. As shown in Fig. 2, the incident X-ray beam from Ge(880) contains high-intensity low-order reflections from Ge(220), Ge(440), and Ge(660), as well as Ge characteristic lines. Therefore, it is possible that the PC detector became saturated with a thin filter in a case of the Ge(880) monochromator crystal, which likely resulted in a reduced count rate. If Fe characteristic lines were also introduced under these conditions, the count rate could be further reduced. This effect likely caused the negative peak observed for μt.

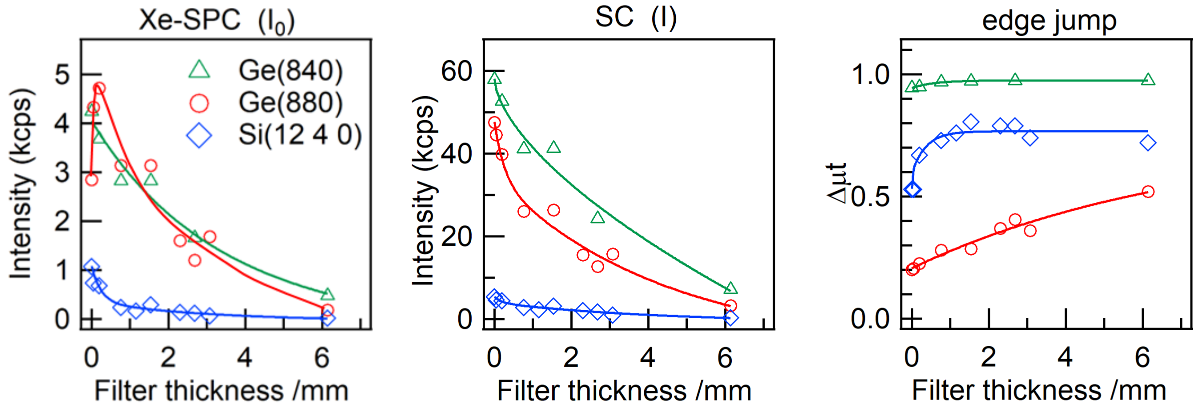

Fig. 4. Dependence of detected X-ray intensities at 25.4 keV on Al-filter thickness recorded with two different detectors and a monochromator crystal, and the resulting difference in the observed edge jump at Ag K-edge. Sample: 20 μm Ag foil.

The edge jumps at the K absorption edge of the same sample should all be the same. However, as seen in Fig. 4, the absorption edge jump of 20 μm Ag foil using the Ge(880) monochromator are clearly smaller than those recorded with the Si(840) and Si(12 4 0). We presume this discrepancy arises from insufficient monochromaticity of the incident X-ray beam. By reducing the undesired X-ray intensity with insertion of Al filter, μt is expected to approach the theoretical value of 1.0. When using the Ge(880) monochromator crystal, the 25.6 keV target X-ray is contaminated with X-rays at 6.4, 12.8, and 19.2 keV originating from low-order reflections. The characteristic X-ray line of Ge lies at 10-11 keV. The attenuation effect of the Al filter is limited because of its absorption coefficient, particularly for 19.2 keV X-ray from the Ge(660) reflection. If X-rays at these undesirable energies also contributed to the measured X-ray absorption spectrum, the resulting edge jump will be underestimated. We presume this is the reason for the smaller edge jump observed when using Ge(880) compared with other two analyzing monochromator crystals used in this study. In such cases, the spectral shape might be broadened. Therefore, careful data-handling procedures are required, particularly for quantitative analysis.

As well-known and discussed [3, 20], the use of a monochromator crystal with smaller d-spacing improves the energy resolution. Fig. 5 shows Pb L3 XANES spectra of PbO_2 recorded using four-kinds of monochromator crystals. The first derivatives are also shown in the Figure. The receiving slit and the divergent slit had different widths. Note the width of each slit was kept constant for the four experiments (0.05 mm). As predicted by Bragg's equation, the energy resolution improved with decreasing the d-spacing. The fine structure of XANES, indicated by arrows, became distinctly observable under these conditions. Notably, the maximum energy value obtained using the Si(400) monochromator crystal differed from those observed with others.

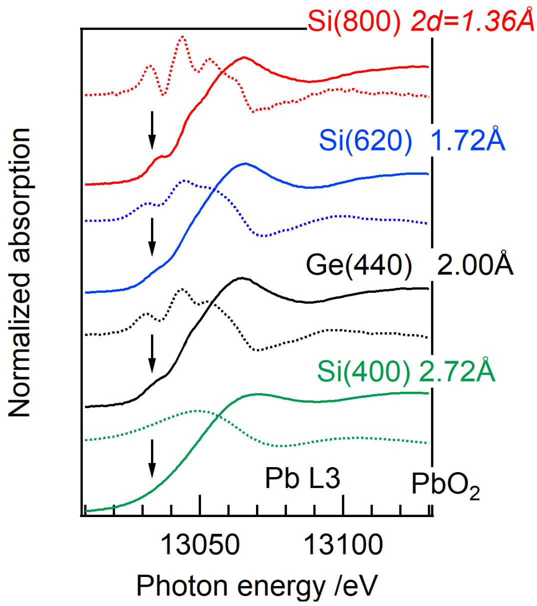

Fig. 5. Pb L3 edge XANES spectra of PbO_2 and the first derivatives recorded with four kinds of monochromator crystals.

Fig. 6 shows Pb L3 edge XANES spectra of Pb compounds recorded with Si(800), Si(620) and Si(400) monochromator crystal. XANES spectra recorded using Si(620) allowed for clear differentiation among each compound. Furthermore, the use of Si(800) gave an energy resolution comparable to those recorded at synchrotron radiation facility SPring-8[21]. Note that when Pb L3-edge X-ray absorption experiment was performed using a Si(800) monochromator crystal, Mo Kβ3 line (19.59 keV) arising from Si(12 0 0) reflection overlapped with the XANES region around 13.06 keV. The incident X-ray intensity obtained using the Si(400) monochromator crystal was more than 10 times larger than that obtained with the Si(800) or Si(620) monochromator crystals. Then, conditions for X-ray generator, the sample-slit (SS) width, and the acquisition times are not the same for the three kinds of experiments.

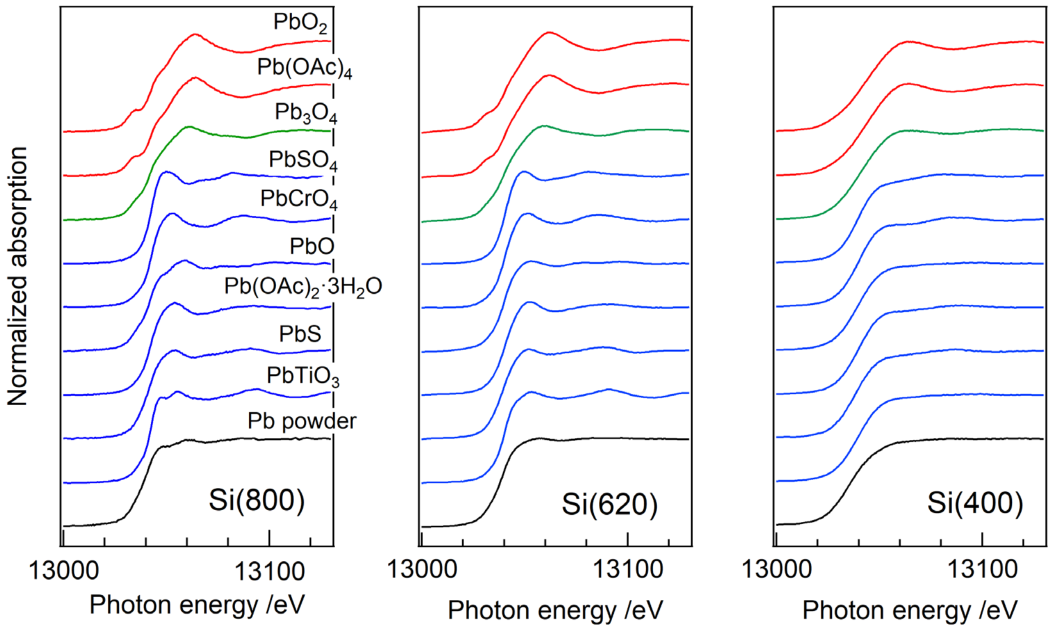

Fig. 6. Pb L3 edge XANES spectra of Pb compounds with different oxidation number and the coordination environment using three types of monochromator crystals. Conditions: Si(800): 21 kV, 50 mA, 150 sec/pt, SS = 15 mm; Si(620): 27 kV, 50 mA, 100 sec/pt, SS = 15 mm; Si(400): 27 kV, 20 mA, 30 sec/pt, SS = 2 mm.

XANES spectra have been widely utilized for oxidation state analysis of unknown samples using the absorption edge. Next, we estimated the apparent absorption edge of each compound to survey influence on the spectral resolution. It is not easy to experimentally estimate the true absorption edge energy from XANES spectra. Although various methods have been utilized for evaluating the apparent edge position [22], no established guidelines are currently available.

Then, the energy value at half of the normalized XANES (E0-half) and the energy value of the peak in the first-derivative (E0-diff) were collected for Pb L3-edge XANES spectra shown in Fig. 6, as a widely utilizing simple and less arbitrary method. Deviations from the zero-valent lead E0 value were summarized in Fig. 7 as a function of oxidation number of each compound. As well known, the edge energy shift to higher energy with increasing the oxidation number of a compound. The slope represents the apparent chemical shift per oxidation number. The slope of E0-half was nearly identical for Si(800) and Si(400), and did not differ significantly from SR study [23, 24] and our previously reported value for Si(620) (1.26 ± 0.06)[25]. However, the E0-half values for Pb^{2+} compounds exhibited a relatively large deviation in the case of Si(800). Pb_3O_4 is a mixed-valence compound containing both Pb^{2+} and Pb^{4+} ions, and the average oxidation number is +2.67. The E0-half of Pb_3O_4 existed within the range of Pb^{2+} compounds, indicating that the edge position in high-resolution XANES alone is insufficient to determine the oxidation state of unknown samples.

The E0-diff values of Pb^{2+} compounds measured on Si(800) were confirmed to overlap with those of Pb^{4+} compounds, indicating that the E0-diff value obtained from the high resolution XANES is not sufficient to determine the average oxidation state of unknown samples. In contrast, the E0-diff plot obtained using Si(400) exhibited a linear correlation. Several peaks due to transition to unoccupied states are overlapping with XANES. The fine structure around the absorption edge influences the spectral shape of the first derivative of XANES. The deviation of E0-half for Pb^{2+} compounds on Si(400) was smaller than that on Si(800). The linearity of the E0-half plot was somewhat higher for Si(400) than those for Si(800).

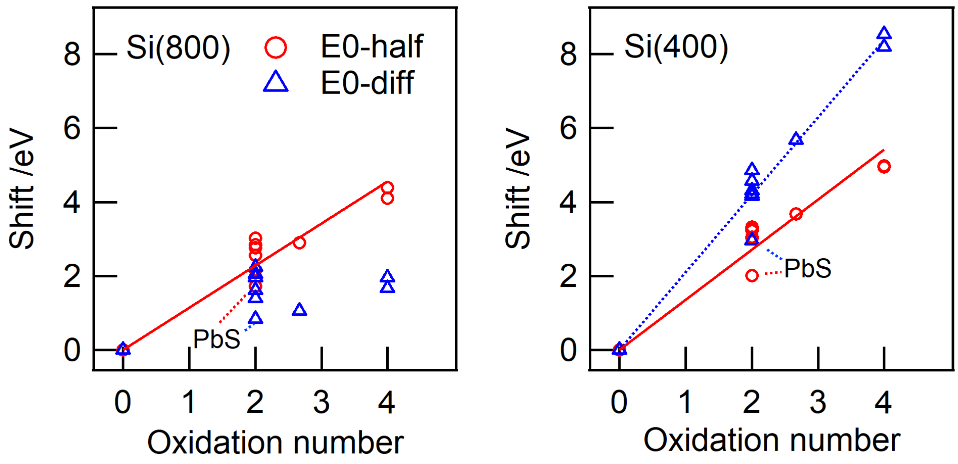

Fig. 7. Apparent absorption edge shift at the Pb L_3 edge from Pb metal, using different monochromator crystals and evaluation methods. E0-half: Energy at half of the normalized XANES. E0-diff: energy at the peak of the first derivative spectrum.

Note we have confirmed clear linear relationship in the E0-half plot of Eu^{2+}–Eu^{3+} mixed simulated spectra at the Eu K-edge, but not at the Eu L_3-edge [26]. The spectral resolution of Eu K-edge XANES is significantly lower than that of the Eu L_3-edge, resulting from differences in the natural width of the core level. The previous Eu K- and L-, and the present Pb L-edge XANES analysis suggest XANES spectra having lower energy resolution would provide more accurate information about the averaged oxidation number of unknown samples. The Pb L_3-edge XANES analysis demonstrates that spectra recorded using Si(800) are suitable for quantitative analysis. In contrast, first derivative spectra obtained with Si(400) are more appropriate for evaluating the average oxidation state of unknown samples. The X-ray intensity obtained using Si(400) is more than 1000 times higher than that with the Si(800) monochromator, allowing measurements to be performed in a shorter time or at lower XG power. As the general tendency, we assume that using X-rays in a low-energy-resolution optical system would allow the average oxidation state to be estimated more accurately and in a shorter time.

Fig. 8 shows Br K-edge XANES spectra of KBr recorded using Si(620) and Si(800) monochromator crystal. The k^3-weighted EXAFS spectra recorded with a wider receiving slit width and the Fourier transforms are also shown. The spectral shape differences observed with Si(800) and Si(620) monochromator crystals were significant, where the spectral resolution of the XANES for Si(620) was drastically broadened. This tendency was different from Pb L3 edge XANES. The natural width at the Br K level (2.52 eV) is approximately half that of the Pb L_3 level (5.81 eV) [27]. The energy of Br K-edge is close to that of the Pb L_3-edge within 500 eV. It suggests that difference of each optical resolution between the two absorption edge is negligible in cases the same monochromator crystal was used. We presume that the instrumental resolution provided by the Si(620) monochromator around 13–14 keV is narrower than the natural width of the Br K-level, but wider than that of Pb L3-level. The energy resolution is generally considered as less influence on EXAFS oscillations. Then the amplitude reduction of Br K-edge EXAFS recorded using Si(620) monochromator was minimal above 4 $Å^{-1}$.

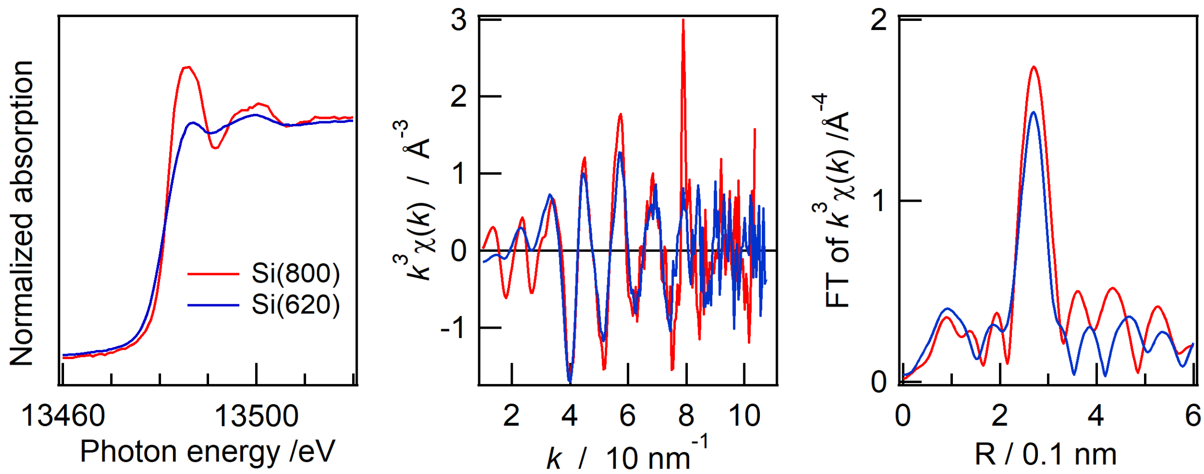

Fig. 8. Br K-edge XANES/EXAFS spectra of KBr with two types of monochromator crystals. Si(800): 80 s/pt (XANES), 600 s/pt (EXAFS); Si(620): 50 s/pt (XANES), 180 s/pt (EXAFS).

Finally, Ag K-edge XANES experiment was conducted. Fig. 9 shows Ag K-edge laboratory and SR XANES spectra. The energy resolution of lab-XANES with a Ge(840) monochromator crystal was only sufficient to barely distinguish between different compounds. The use of a Ge(880) monochromator crystal (2d = 1.00 Å), in combination with a divergence slit (DS) width reduced to half that used for Br K-edge and Pb L_3-edge measurements, provided XANES spectra comparable to those obtained at SR facility. The resolution is sufficient to be applicable to chemical state analysis. Although lattice spacing of Si(12 4 0) (2d= 0.857 Å) is smaller than that of Ge(880), the spectral resolution was evidently lower. It has also been established that not only the lattice parameter of a monochromator crystal, but also the absorption coefficient relates to energy resolution of the present X-ray curved-crystal spectrometer [3, 20]. The absorption coefficient of Si at Ag K-edge is one-tenth that of Ge, resulting that the two types of monochromator crystals showed the energy-resolution relationship described above.

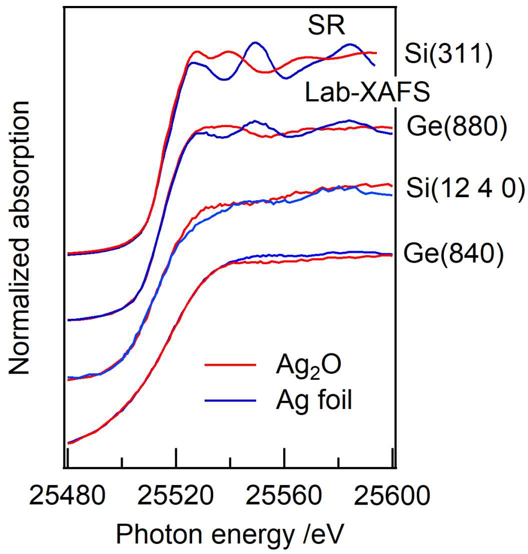

Fig. 9. Ag K-edge laboratory XANES spectra of Ag_2O and Ag foil obtained using various monochromator crystals with different d-spacings. SR XANES was recorded at BL01B1 at SPring-8.

The present XAFS experiment was conducted using a commonly used scintillation counter with a diameter of 1.5 inches as for a transmission X-ray detector. The Ag K-edge XANES measurement using Ge(880) required approximately 12 hours due to extremely low X-ray intensity (Note that the measurement time for each X-ray absorption spectrum shown in Fig. 3 was 2 hours.). Under the current conditions, Ag K-edge EXAFS analysis is not practically feasible because of the low counting rate. We presume replacing the present scintillation counter with a large-area energy dispersive detector (e.g., SDD or strip detector) could improve the count rate and allow energy-selective counting near the target X-ray energy. This would substantially reduce the measurement time, improve the signal-to-background ratio, and potentially enable EXAFS analysis.

4. Conclusion

Pb L_3-, Br K-, and Ag K-edge XANES spectra were recorded using a conventional laboratory-type spectrometer equipped with five monochromator crystals: Si(400), Si(620), Si(800), Ge(840), or Ge(880). To suppress low-order reflections and characteristic Ge K-lines, a thick aluminum filter was inserted into the incident X-ray path.

The Pb L_3-edge absorption of various compounds was evaluated using two methods: E0-half (half-height of normalized XANES) and E0-diff (peak of the first derivative). The energy resolution had no significant effect on the apparent chemical shift derived from E0-half, while it significantly influences the shift determined from E0-diff. Both values showed greater deviation for Pb^{2+} compounds when using the high-resolution Si(800) crystal, suggesting that edge position alone may be insufficient for oxidation state determination without complementary evidence. In contrast, lower-resolution spectra provided more reliable estimates of average oxidation number via first-derivative analysis.

Although the Br K-edge is close in energy to the Pb L_3-edge, the spectral shape differences between Si(800) and Si(620) varied significantly between the two edges. This highlights that optical resolution has a stronger impact on XANES features for edges with narrower natural widths, such as Br K-edge. Additionally, high-order Ge(880) reflections enabled Ag K-edge XANES measurements suitable for chemical state analysis.

Acknowledgements

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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